Physics Test [Electricity]

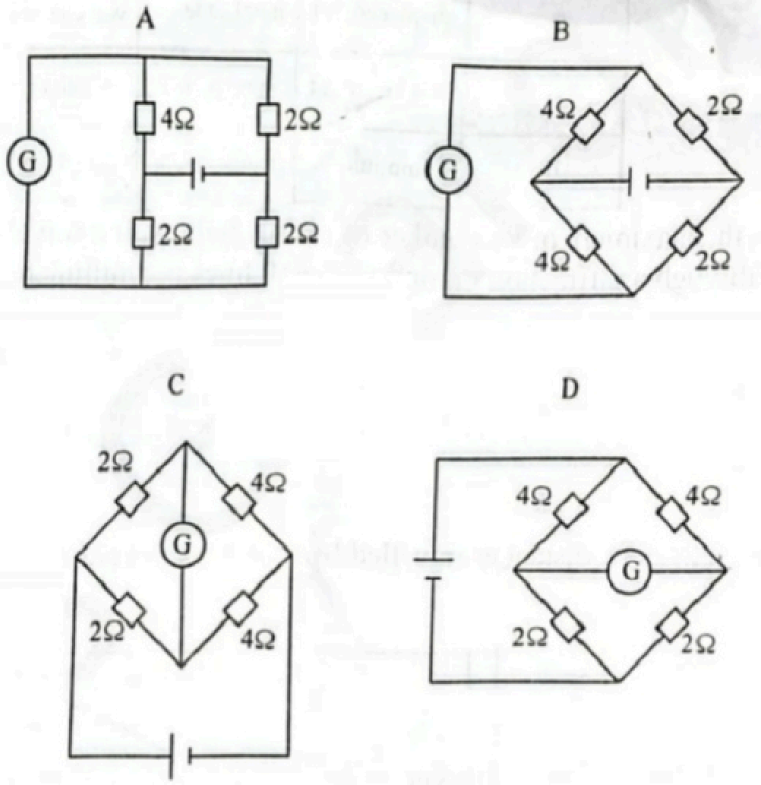
1. What is the advantage of using high voltage for transmission of electrical energy?

   A A steady output voltage is gained.
   B Resistance per unit length is reduced.
   C More current is transmitted.
   D Power loss is reduced in transmission cables.

2. Which one is **not** an application of the electrostatic phenomena?

   A photocopying
   B spray painting
   C filtering
   D dust extraction

3. In which circuit will the centre zero galvanometer show a deflection?



4. When an alternating current flows through a resistor of resistance 12 Ω, heat is dissipated at a rate of 48 W.

   What is the peak value of the alternating current?

   A 4.00 A
   B 2.83 A
   C 2.00 A
   D 1.41 A

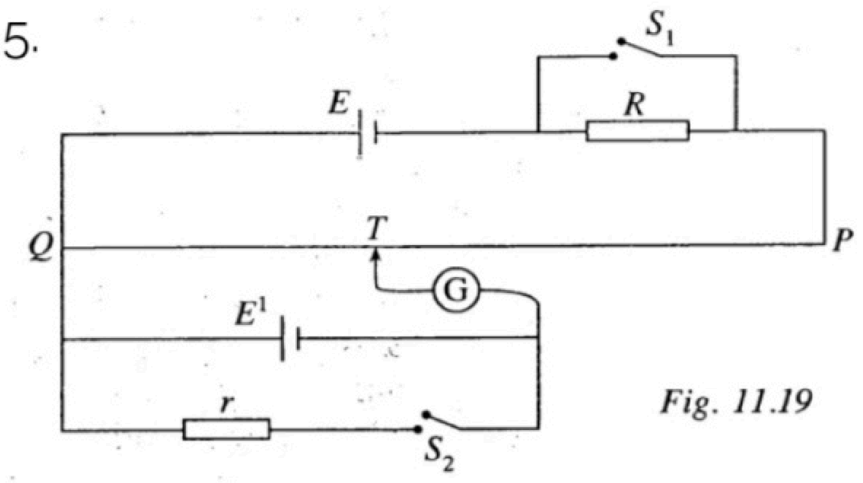
5. 

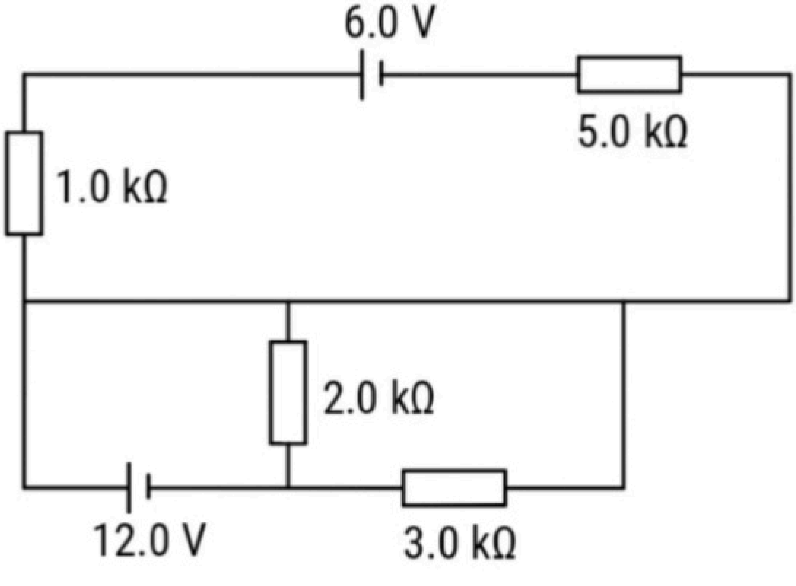
Fig. 11.19

In Fig. 11.19, PQ is a uniform wire of length 1.0 m and has a resistance of 10.0 Ω. E is an accumulator of e.m.f. 2.0 V and negligible internal resistance. The resistor R has a resistance of 15 Ω and r is a 5.0 Ω resistor. With both the switches S_1 and S_2 open, the balanced length QT is 62.5 cm. When S_1 and S_2 are closed, the balanced length is 10.0 cm. Calculate

(i) the e.m.f. of cell E^1,
(ii) the internal resistance of cell E^1,
(iii) the balanced length QT when S_2 is open and S_1 closed,
(iv) the balanced length QT when S_1 is open and S_2 closed.

[7]

6. Calculate the currents flowing through the 5.0kΩ, 3.0kΩ, 2.0kΩ and 1.0kΩ resistors.



[4]

7. The circuit diagram in Figure 10.27 shows a 12 V power supply connected to some resistors.

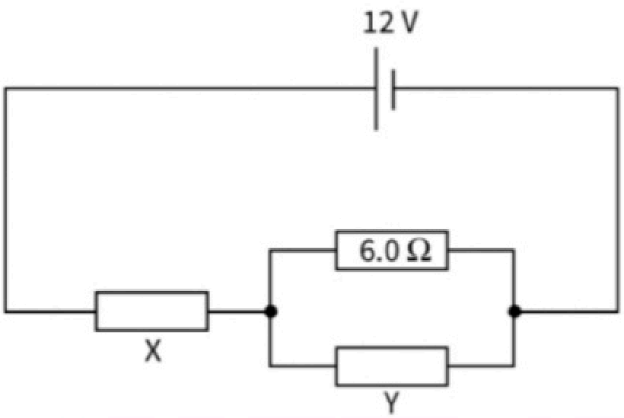


Figure 10.27

The current in the resistor X is 2.0 A and the current in the 6.0 Ω resistor is 0.5 A. Calculate:

a the current in resistor Y

b the resistance of resistor Y

c the resistance of resistor X. [5]

8(a) Define

1. *charge*,

2. *Coulomb*.

(b) Explain how electrostatics is used to extract dust particles from chimneys of factories. [5]

Total : 25